

HARJEET SINGH PANNU

ELECTRONICS ENGINEER • VLSI & SEMICONDUCTOR • EMBEDDED SYSTEMS & AI/ML INTEGRATION

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PROFESSIONAL SUMMARY

B.Tech (Electronics & Communication Engineering) graduate from VIT Chennai (CGPA 9.4/10), certified in VLSI Design & Verification, with hands-on project experience spanning embedded systems, AI/ML, and IoT — including an AI-powered autonomous robot (YOLOv8, OpenCV, Raspberry Pi) and a real-time IoT sensing platform. Ranked #79 of 76,886 worldwide in the WorldQuant Quantitative Championship 2025. Structured technical research experience from a JPMorgan Chase Research Analyst internship (Python, MATLAB, Java, C++). Oracle Certified Foundations Associate in Agentic AI, with hands-on experience using Claude Code to automate real workflows and exploring LangChain. Seeking an entry-level Applications/Product Engineer or Graduate Engineer Trainee role in the semiconductor industry.

PROFESSIONAL EXPERIENCE

JPMorgan Chase & Co. — Commercial & Investment Banking Division

Jan 2026 – Jun 2026

CRG Research Analyst Intern

Mumbai, India

- Conducted structured technical and market research, translating complex data into clear findings for cross-functional stakeholders.
- Analyzed large, multi-source datasets and performance metrics to build accurate, data-backed technical reports.
- Authored structured technical reports and presentations reviewed by senior stakeholders, following consistent documentation standards.
- Applied quantitative research frameworks and Python/MATLAB-based data analysis to track trends and support fact-based engineering decisions.
- Collaborated with cross-functional, global teams in a fast-paced environment, maintaining high accuracy and attention to detail.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

2022 – 2026

B.Tech, Electronics & Communication Engineering

CGPA: 9.4 / 10.0

- Academic Excellence Award — Top Performer, B.Tech ECE

Apeejay School, Kolkata

2021 – 2022

CBSE Class XII

Percentage: 98%

Podar International School, Aurangabad

2019 – 2020

CBSE Class X

Percentage: 96%

SKILLS & COMPETENCIES

Hardware & Embedded: VLSI Design & Verification, Embedded Systems, IoT, AI/ML, Wireless Comms, Generative & Agentic AI (Oracle Certified Foundations Associate; hands-on: Claude Code, LangChain)

Programming & Data: Python, MATLAB, Java, C++

Tools & Platforms: Excel (Advanced), PowerPoint, Git, VS Code, Postman

Research & Analytical: Technical Research, Quantitative Analysis, Structured Documentation, Cross-Functional Collaboration

ACADEMIC & TECHNICAL PROJECTS

IntelliWeed — AI-Powered Autonomous Agricultural Robot [\[Brochure\]](#) [\[Demo\]](#)

- Built an AI-powered weed-removal robot with YOLOv8, OpenCV, and Raspberry Pi — 90% detection accuracy, 40% less labor.

Smart Multi-Biometric UPI Payment Authentication System [\[GitHub\]](#) [\[Patent Status\]](#) (App. No. 202541081764)

- Designed a multi-factor biometric authentication system combining fingerprint and permutation-based entry — cut fraud 40%.

Smart Plant Health Monitoring System — Industrial IoT Platform

- Built a real-time IoT sensing and irrigation platform with Arduino and Blynk — improved water-usage efficiency by 50%.

ACHIEVEMENTS & CERTIFICATIONS

- WorldQuant Quantitative Championship 2025 — Global Rank #79 of 76,886 participants [\[Award Certificate\]](#) [\[Rank Certificate\]](#)
- Academic Excellence Award — Top Performer, B.Tech ECE, VIT Chennai
- 1st Runner-Up, Pitch-A-Thon — competed against 70+ teams [\[Certificate\]](#)
- International Research Finalist — ICDTSA Conference [\[Certificate\]](#)
- Certified in VLSI Design & Verification [\[Verify\]](#) (Code: FRFGXK4K)
- Oracle Certified Foundations Associate — Agentic AI [\[Verify\]](#) (Aug 2026)

LEADERSHIP & EXTRACURRICULAR

Sponsorship & Outreach Lead — Fraternity of Young Innovators (FYI)

- Led industry outreach and corporate sponsorships for the club, developing stakeholder management and negotiation skills.